

Food chains

ENERGY PASSES ALONG FOOD CHAINS FROM PLANTS TO TOP CARNIVORES.

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Living things require a supply of energy and nutrients to power, maintain, and grow their bodies. Scientists track how energy and nutrients move from one organism to another using food chains.

Producers and consumers

Food chains always begin with plants and other photosynthetic organisms, which are known as the producers. Animals and other heterotrophs (organisms that eat others to survive) are known as the consumers. The nutrients and energy gathered by the producers passes up the food chain via a series of consumers.



△ Producer

Green plants harness the energy of sunlight to power themselves, and are called producers.



△ Primary consumer

Herbivores, such as cows, eat only producers, and form the second step in the food chain.



△ Secondary consumer

Omnivores, such as raccoons, eat both producers and small primary consumers.



△ Top predator

The food chain ends with a powerful predator, such as an eagle or shark.



△ Detritivore

Worms, vultures, and most fungi recycle the dead remains and waste of other organisms.

Energy pyramid

Most of the energy consumed by organisms is given off as heat, becoming unavailable to the rest of the food chain, so less energy is passed onto the next level. As a result, the total quantity of organisms—the biomass—also decreases. This gives the food chain a pyramid structure—with many producers at the base, and fewer and fewer consumers at each stage above.

▷ Trophic levels

Scientists call each level of a food chain a trophic level—from the Greek word for food. As a rough estimate, only about 10 percent of the energy in one trophic level passes to the one above.

